SULAV GAIRE

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Sulav-Gaire | SulavGaire

Bhaktapur, Nepal

Research Interests

My research interests focus on Electronics, Machine Learning, and IoT, with particular emphasis on **mobile computing and cyber-physical systems**. I am **currently researching Wi-Fi sensing** in a personal project and aim to further develop my work in these fields.

Experience

- **Nepal Academy of Science and Technology (NAST)[]** — *Dec 2024 - Present*
  Innovative Research Assistant — Khumaltar, Lalitpur, Nepal
  - Developing wireless sensor network for early waring system
  - Researching tomato picking robot using robotic arm and imitation learning.
  - Handel electronics aspects of projects like hydrogen fuel cell, GPS data logger.
- **ORION Space []** — *Feb 2022 - Aug 2022*
  Electronics Design Intern — Duwakot Mod, Bhaktapur
  - Researched LoRa-based sensor networks in UHF band
  - Engineered educational satellites (CanSat, CUBEEK) used in global space training programs [20+ countries]
  - Developed web application for real-time LoRa data monitoring

Education

- **Institute of Engineering, Tribhuvan University** — *Nov 2019 – April 2024*
  Bachelor's Degree — Pokhara, Nepal
  - Electronics, Communication and Information Engineering

Projects

- **StaticSense: Static Object Detection using Wi-Fi Channel State Information** — *2025*
  Tools: Resnet, Signal Processing, Esp32-WiFi-CSI []
  - Researched static object detection using Wi-Fi CSI, achieving 98.27 percent accuracy, addressing a challenge previously limited to dynamic activity sensing. Accepted in **IEEE International Conference on ICT and Photonics (ICTP), 2026**
- **CUBEEK & EDUCAN: Educational Satellites** — *2022*
  Tools: Node.js, Arduino, PCB Design, LoRa []
  - Engineered 1U CubeSat kit with solar power systems and LoRa communication modules
  - Developed real-time dashboard and optimized PCB design, reducing production costs by 30%
- **Electronic Voting System: Biometric Authentication** — *2024*
  Tools: Python, Flask, OpenCV, SHA256 Hashing []
  - Developed secure two-factor biometric voting system with fingerprint and facial verification
  - Achieved positive user ratings from 50 test participants with intuitive, interactive interface
- **Waste Classification Delta Arm** — *2024*
  Tools: YOLOv8, OpenCV, Arduino []
  - Built AI-powered sorting robot achieving 94.1% precision, winning 1st place among 50 teams
  - Engineered end-to-end system for classifying waste into four categories with delta arm mechanics
- **High Altitude Weather Forecasting** — *2023*
  Tools: LoRa, XGBoost, Python, Arduino []
  - Created cost-effective LoRa-based weather prediction system for remote mountainous regions
  - Implemented XGBoost algorithm to forecast patterns from sparse environmental sensor data

Publications

C=Conference, J=Journal, P=Patent, S=In Submission, T=Thesis

[C.1] Shiva Shrestha, Samman Shrestha, Prateek Paudel, Sudarshan Gurung, Sulav Gaire, Smita Adhikari (2024). **Deep Learning for Waste Management: Leveraging YOLO for Accurate Waste Classification**. In *Proceedings of the 15th IOE Graduate Conference*, Volume 15, May 2024. Institute of Engineering, Tribhuvan University. ISSN: 2350-8914 (Online), 2350-8906 (Print).

SKILLS

- **Programming Languages:** Python, MATLAB, JavaScript, Shell, C, C++
- **Web Technologies:** MERN Stack (MongoDB, Express.js, React, Node.js)
- **Database Systems:** MongoDB, MySQL
- **Data Science & Machine Learning:** Pandas, NumPy, Matplotlib, TensorFlow, PyTorch
- **Specialized Area:** IoT, Low-Power Wide-Area Networks (LPWAN), Embedded Systems
- **Other Tools & Technologies:** KiCad, LTspice, Figma, ROS2

HONORS AND AWARDS

- **Poster Presenter – WiFi-HAR: Activity Detection Using Channel State Information** *2025*
  International Conference on Science and Technology Information Management System: Practices and Experiences
  - Organized by NAST and NAM S&T CENTRE)
- **National 2nd Runner-up – TECHNERGY Hackathon** *2024*
  Hitachi Energy []
  - Recognized among top 3 out of 20+ national teams for software testing
- **1st Place – Project Expo (Delta Arm)** *2024*
  NebiFest 2024, IOE WRC
  - Ranked 1st among 50+ projects for developing a low-cost robotic arm with real-time control
- **Regional 2nd Place – Robo-Soccer** *2021*
  CAN-INFOTECH Robotics Competition

LEADERSHIP EXPERIENCE

- **Senior Member – Robotics Club** *2020 – 2023*
  IOE, WRC
  - Coordinated a 15-day robotics training for 100+ first-years freshers
  - Led organizing team for in-house robotics competition; 20+ teams participated
  - Mentored juniors in embedded systems and Arduino-based prototyping

ADDITIONAL INFORMATION

Languages: English (Proficient), Nepali (Native Speaker), Hindi (Intermediate)
Interests: Engineering, Art, Dance

:) ***HAPPY ENGINEERING***